HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Configured and integrated new software-controlled laboratory systems with **vendor APIs** and internal databases, ensuring alignment with GMP and regulatory compliance.
- Developed Python scripts to automate data validation checks for laboratory system configurations, reducing manual error rates by an estimated **15%**.
- Managed system documentation lifecycles, including creation and revision of SOPs and validation protocols for laboratory instrumentation, supporting audit readiness.
- Collaborated with **IT, vendors, and lab users** to define system requirements and integration points for new laboratory software, facilitating smooth project delivery.
- Supported the implementation of data integrity controls within laboratory systems, ensuring data traceability and audit trails in line with regulatory expectations.
- Performed root cause analysis on system integration issues for laboratory equipment, implementing corrective actions to maintain operational stability.
- Containerized laboratory system integration services using **Docker** for consistent deployment across development and production environments.

SKILLS

Domain Expertise – GMP, Data Integrity, Regulatory Compliance, Laboratory Systems, Quality Control, Process Automation.

Programming Languages – Python, SQL, Java.

Databases – PostgreSQL, MySQL, Oracle.

Cloud & Containerization – AWS (EC2, S3, Lambda), Docker.

Scripting & Automation – Shell Scripting, Cron Jobs.

Development Tools – Git, Jenkins, Jira, Confluence.

Data Visualization – Power BI, Tableau, DAX, Power Query.

PROJECTS

Automated Laboratory Data Validation Service

- Developed a Python-based service to automate the validation of data outputs from integrated laboratory systems against predefined quality rules.
- Implemented data extraction and transformation logic using **pandas** and **SQL** to process ~5,000 records daily from disparate lab sources.
- Designed a REST API using **Flask** to expose validation results, allowing downstream systems to consume integrity reports.

Lab System Integration & Configuration Management

- Engineered a configuration management framework using Python scripts and **SQL** to standardize the setup of new laboratory instruments.
- Created detailed technical documentation and SOPs for system integration procedures, reducing onboarding time for new lab equipment by **20%**.
- Collaborated with lab operations teams to gather requirements and perform user acceptance testing for integrated systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)